

Moolya Testing

Test Automation for Industrial Automation Systems

Proven Success and Scalable Approach

■ 14+

Years in Business

■ 200+

Customers

■ 20+

Countries

■ 80%

Industry leaders choose us



Who we are

At Moolya Software Testing Private Limited, we lead globally in software testing and quality assurance, partnering with clients across industries to deliver top-tier solutions. Our services ranging from functional and automation to performance and security testing are designed to enhance software quality, efficiency, and productivity, ensuring client success and satisfaction.

Why Choose us



Skin in the Game

We go beyond vendor roles, fostering deep co-ownership with customers.



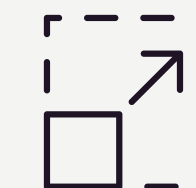
Culture Focus

We help customers address issues by focusing on cultural fixes.



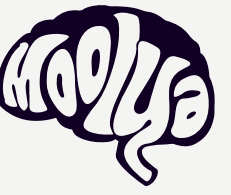
Leadership Friendly

Moolya empowers C-, V-, and D-level leaders with insights.



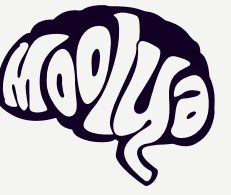
Scaling Value

Moolya unifies Tech, Test, Product, Customer & Business insights to scale value.



Moolya Growth Driven Testing framework

S:No	Aspect	Growth Driven Testing	Regular Testing
01	Approach	Adapts methodologies based on project context.	Follows predefined testing processes agnostic to projects.
02	Flexibility	Flexible in choosing techniques and approaches.	Testing procedures are not changed and are prefixed.
03	Emphasis	Focuses on critical thinking and exploration.	Emphasizes adherence to test plans.
04	Testing Goals	Prioritizes on testing being aligned to Business goals and moving from Bug finding to prevention of bugs.	Aims to verify compliance with requirements.
05	Mindset	Encourages creativity and adaptability.	Emphasizes structured and procedural approaches.

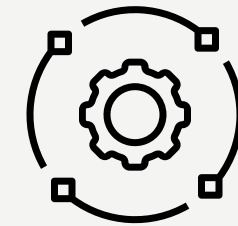


Solutions **Led Delivery**



360° Testing Solution

Holistic testing for comprehensive coverage and risk mitigation, enhancing quality and satisfaction.



Fastrack Automation

Dynamic, tool-agnostic automation to reduce time-to-market and ensure thorough test coverage.



Rapid Deployment Testing Solution

Streamlined testing for faster release cycles and confident, quality product delivery.



SpeedMaster Optimization Solution

Performance optimization to boost application speed, responsiveness, and scalability.

Advantages of Solution-Led Delivery

Alignment with Customer's

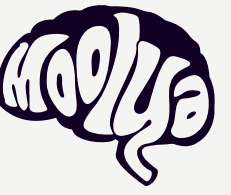
Digital transformation goals

Testing Solutions

Address unique challenges & requirements

Accelerated

Time to Market & Reduced Cost



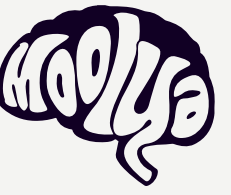
Fastrack Automation

What is it?

Fastrack Automation, an advanced solution from Moolya, employs intelligent techniques to expedite your automation process in the existing environment without any platform lock in half the time.

Fastrack Automation optimizes test case design, execution, and maintenance. The service dynamically adapts to changing application landscapes, ensuring sustained efficiency gains.





Fastrack Automation

Leveraging Technology Platform and Expert Human Insight for Intelligent Automation

Profile

Design Tests

Drag & Drop, Wizard driven, Easy-to-use, Domain-Intuitive Application Profiling.

Feature Files

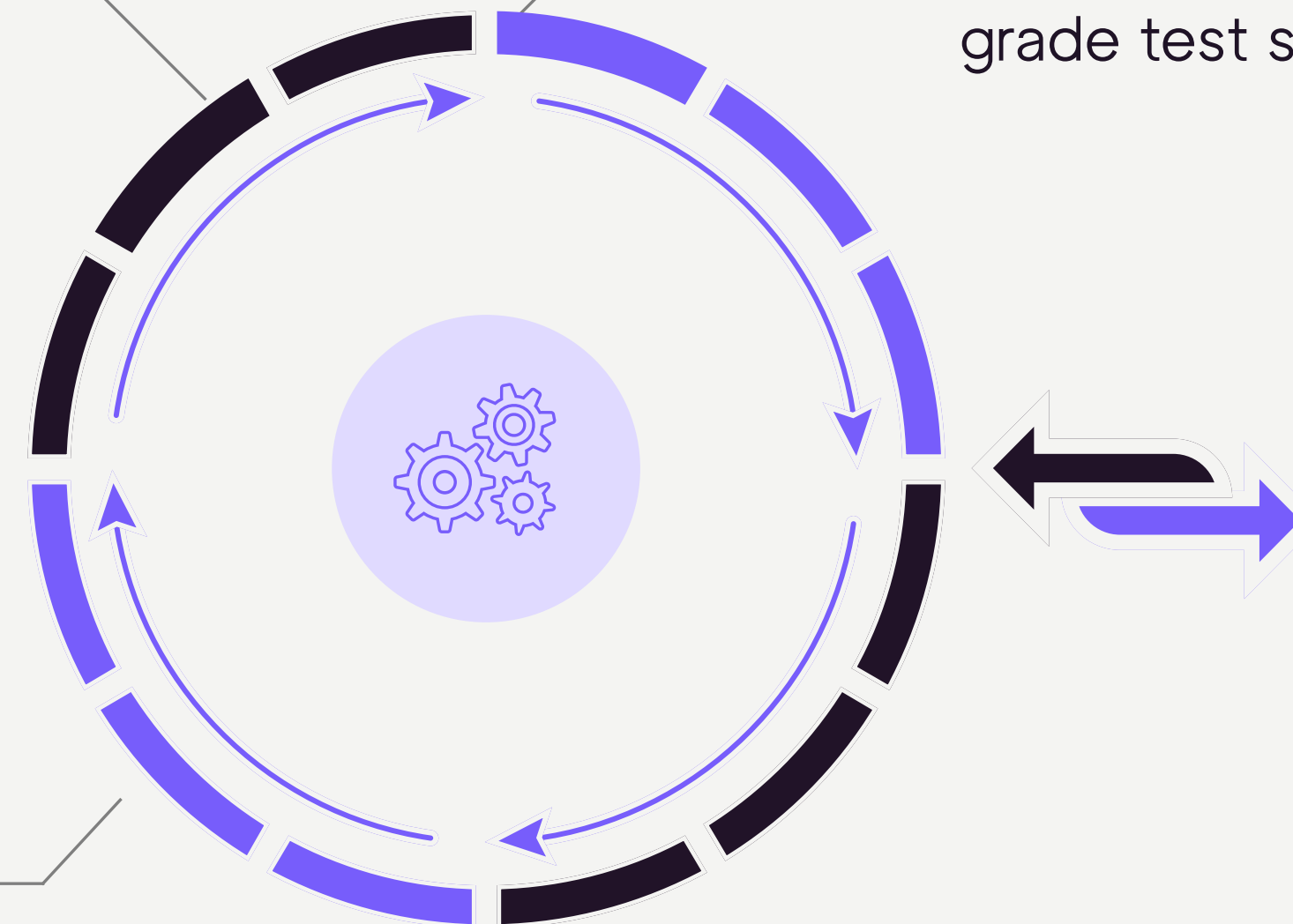
Auto-generation of optimal number of test cases written in Gherkin language, along with datasets.

Schedule & Execute

Schedule Test Execution, Collect Results, Publish Reports & Dashboard.

Auto-Code

Auto-Generation of production-grade test scripts.



Moolyan - Experienced software tester skilled in crafting robust test automation scripts.



Fastrack Automation

End-to-End automation platform powered by **Artificial Intelligence**.

Abstraction is our differentiator

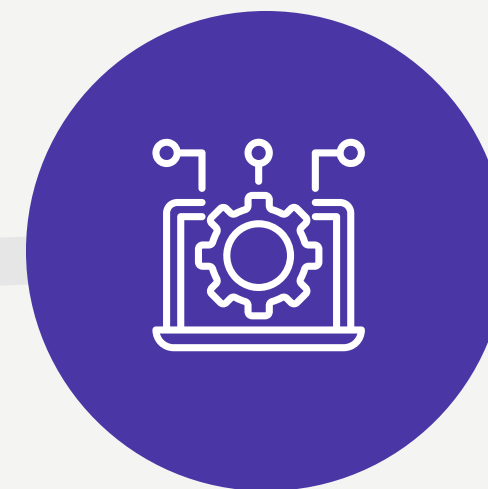
One Platform

- Handles UI Based, API, and Load Test Automation.



Technology Agnostic

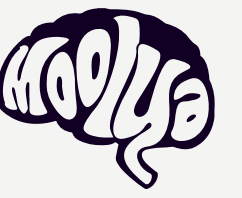
- Supports multi-language scripting.
- Seamless working across ALMs.
- Multiple reporting formats to choose from.



All Types of Applications

- Web, Desktop, Mobile, and Hybrid Applications.
- Robotic Process Automation.
- Embedded Software.





Fastrack Automation – Platforms & Tools supported for Web, Mobile and Desktop

Application types

Desktop

Mobile

Web

Hybrid

Test Automation

UI

API

Load

Embedded

Reporting Formats



Allure



Extent



Cucumber



HTML



XML



JSON



PDF



CUSTOM



WORD

ALM Tools



Jira



TFS

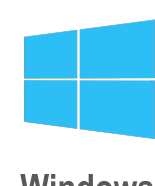


HP ALM



TestRail

Operating System



Windows



Linux



MacOS

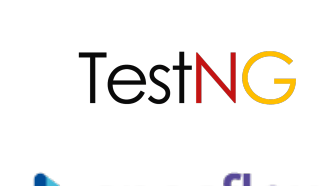
Execution Engine



Behave



nUnit



TestNG



SpecFlow



PyTest

DevOps Tools



Circle CI



Jenkins



GitLab



GitHub

Execution Cloud



BrowserStack



LambdaTest



Kobiton



Azure



AWS

Languages



Java



JavaScript



TypeScript



Python



C#

Framework



FLA UI



Protractor



TestComplete



Flutter



Selenium



Appium



LeanFT



WebDriverIO



Karate



Playwright



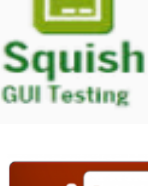
Cypress



RestAssured



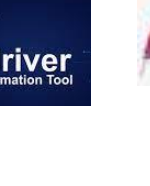
RestSharp



Squish



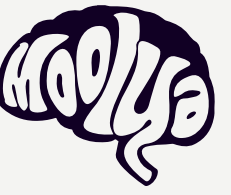
WinAppDriver



Apache JMeter

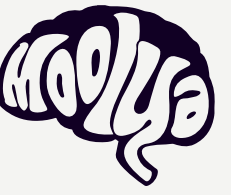


Sikuli



Why Fastrack Automation

S:No	Aspect	Fastrack Automation	Regular Automation
01	Scripting	Fastrack Automation utilizes intelligent techniques such as AI and ML to automate test case creation, execution, and maintenance.	Traditional test automation often involves manual scripting of test cases using programming languages or automation tools.
02	Platform	Fastrack Automation is designed to be platform-agnostic, allowing testing across a wide range of technologies and environments without lock-in.	Automation scripts may be specific to certain platforms or technologies, leading to platform lock-in.
03	Maintenance	Fastrack Automation automates maintenance tasks, such as updating test scripts in response to changes in the application, reducing manual effort and improving reliability – Self Healing.	Maintenance of automation scripts requires continuous effort, especially when dealing with changes in the application under test.
04	Scalability	Fastrack Automation offers a scalable architecture that can handle large-scale test automation efforts with ease, enabling teams to efficiently test complex applications.	Scaling automation efforts to handle large test suites or complex applications can be challenging and may require additional resources.
05	Setup	Fastrack Automation provides a quick and easy setup process, allowing teams to start automating tests rapidly without extensive configuration.	Setting up automation frameworks and environments can be time-consuming and requires technical expertise.



Moolya's Approach to Fastrack Automation

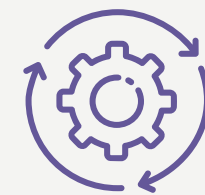


Identify Test Scenarios



Creating Manual Test Cases

Using Scaffolding Framework.



Create Automation Test Cases

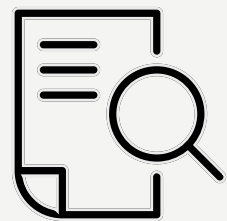


Run Automation

Append the script Regression.

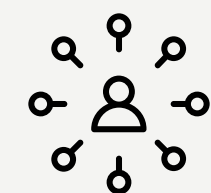


Moolya's Scaffolding Framework



Context Understanding

Understanding the requirements with Business impact, Technical implementation, Why implement it now , How are you managing now without this requirement, Existing automation approach, Current challenges in automation.



Influencing

Tag relevant stakeholders and provide test automation reports including the test coverage details.



Strategising

Focus on user scenarios and functionality during testing. Manual testing of complex workflows is time-consuming. Build a scalable, maintainable automation framework to cut down manual effort, saving time and cost by choosing the right tools."



Performing

In test planning, we review smoke charters, create test cases, and script feasible ones for automation.



Fastrack Automation – Fitment

Greenfield Situation

- New product implementation.
- Testing is predominantly manual.
- SAASification or going mobile.

Legacy Applications

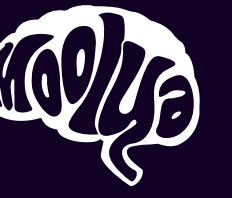
- Applications that have not had testability considerations .

Brownfield Situation

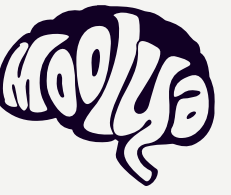
- Replace deprecated/slow tools (e.g., Protractor) with faster, scalable alternatives.
- Move from proprietary to open, cost-effective frameworks.
- Enhance existing frameworks to accelerate automation and reduce flakiness.

Distributed Testing

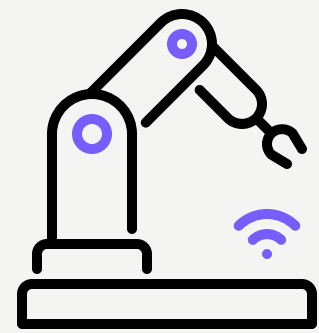
- Parallel Execution.
- Coordinated Parallel Workflows.
- Workflow cuts across Multiple Applications.



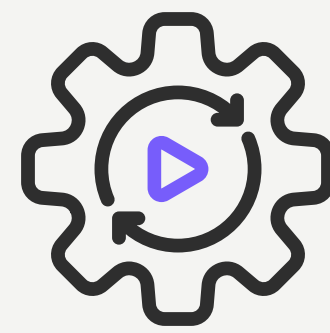
Test Automation for DCS System



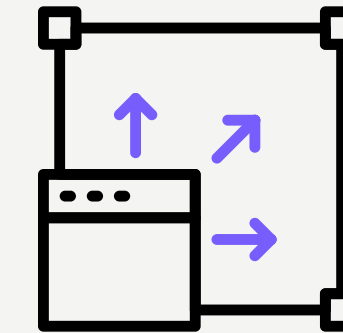
Case Study – Enhancing Reliability through Scalable Test Automation



The customer operates in the **Industrial Automation / DCS** domain where system stability and reliability are paramount.



They had a **basic automation setup** using TestComplete with VBScript.



Their primary goal was to establish a structured and scalable **Smoke and Regression** test suite to support their frequent software updates and configurations.



Challenges Faced by the Customer



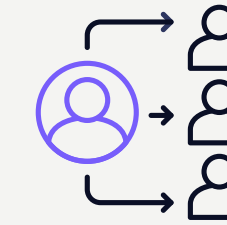
Delays in Releases

Manual and slow testing cycles delayed product releases and upgrades.



High Cost of Maintenance

Legacy VBScript framework required high manual effort for script updates.



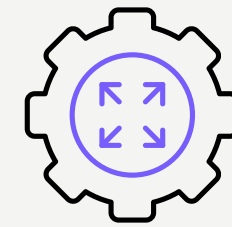
People Dependency

Only a few individuals understood the legacy framework. Risk of knowledge silos and lack of continuity.



Poor Visibility and Coverage

No clear mapping between manual test cases and automation. Gaps in coverage led to production defects.



No Scalable Execution

Test runs not aligned to schedules; no ability to run in parallel or at scale.

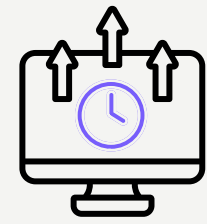


Inconsistent Reporting

Lack of actionable, detailed test reports delayed defect triage.

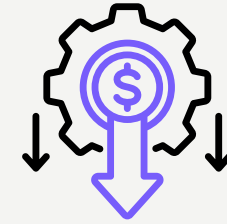


Our Solution Overview



Addressing Delays in Releases

Prioritized automation of smoke test cases to enable early validation of critical workflows, providing faster feedback and reducing release cycle time.



Reducing High Cost of Maintenance

Replaced the legacy VBScript-based framework with a modern TestComplete + Python solution that is easier to maintain and enhances scripting efficiency.



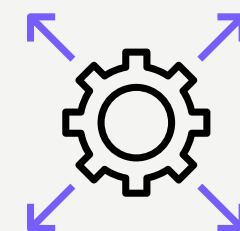
Limiting People Dependency

Designed a simplified, modular framework and provided a low-code/no-code interface that allowed a small team to contribute to automation, reducing reliance on a large scale team.



Improving Visibility and Coverage

Established traceability between manual test cases and automated scripts to ensure better coverage and reduce the risk of production defects.



Enabling Scalable Execution

Set up distributed, multi-system execution for daily test runs, allowing the framework to scale with application complexity and testing demand.

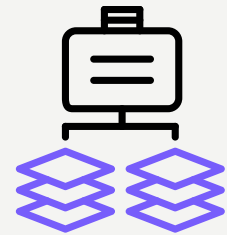


Delivering Actionable Reporting

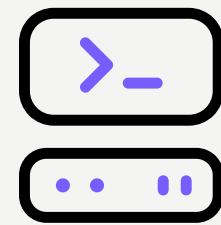
Implemented detailed and structured reporting formats (HTML/PDF) that enabled quicker defect triage and informed release decisions.



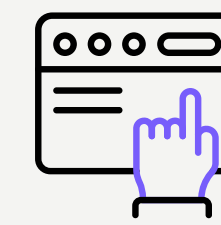
Framework Architecture



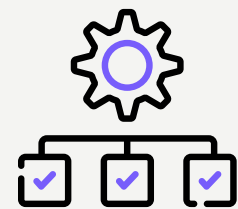
Developed a **modular and layered architecture** using **TestComplete** as the core tool with Python for scripting.



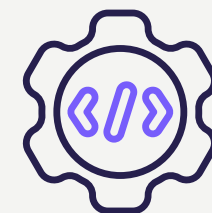
Added a **low-code interface** on top for ease of test creation (e.g., form-based input, test case importers).



Designed the framework to support **mapping between manual test cases and automated scripts** via test case IDs.



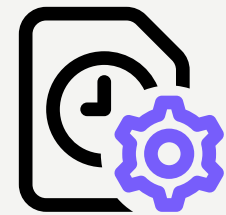
Configured **multi-system execution** setup for parallel and distributed test runs.



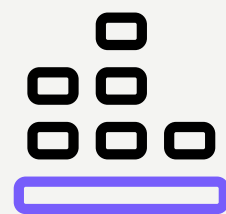
Prepared the framework to integrate with **CI/CD systems** (Jenkins, Github Actions, etc.) when required in the future.



Execution & Reporting Strategy



Set up **daily scheduled test executions** across multiple virtual machines or test beds to ensure consistent validation.

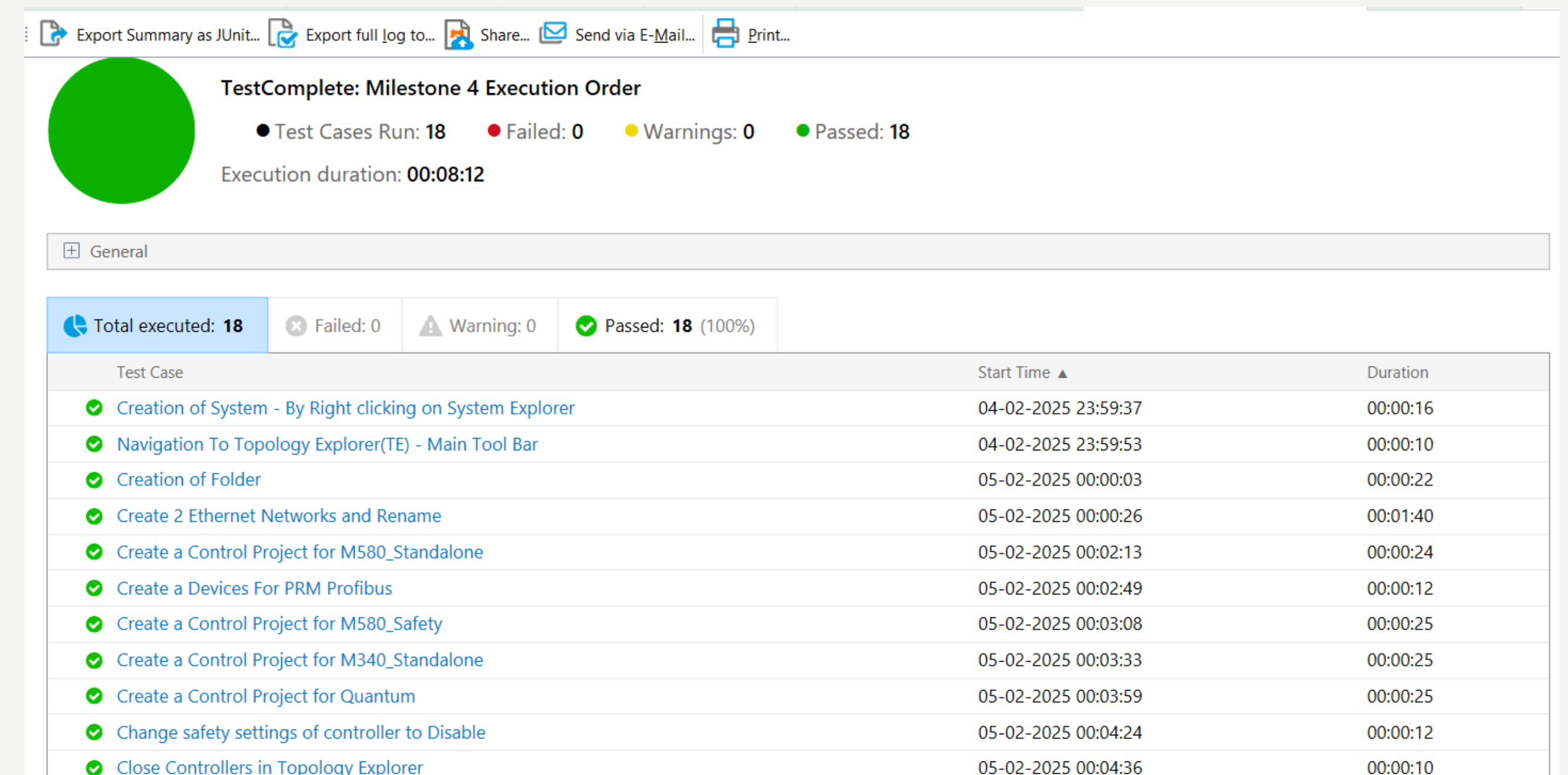


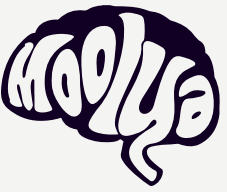
Created an **execution matrix** that allows running specific suites (Smoke, Regression, Feature-wise).



Conducted **weekly demos and review sessions** with stakeholders to showcase progress and gather feedback.

Sample Test Report in TestComplete





Sample PDF Report

Test Execution Summary		
TestComplete: EX-1 ● Test Cases Run: 257 ● Failed: 0 ● Warnings: 1 (0%) ● Passed: 256 (100%) Execution duration: 01:59:26		
Test Case	Start Time ▲	Duration
✓ System Server Launch	04-03-2025 10:17:31	00:00:05
✓ Enter Valid password and Login in SS	04-03-2025 10:17:37	00:00:30
✓ click on Basic Settings in SS Console	04-03-2025 10:18:07	00:00:01
✓ Open hosting in basic settings in SS	04-03-2025 10:18:08	00:00:03
✓ CE tool slot management - Enter 4 Instance of CE and start the system server	04-03-2025 10:18:11	00:01:23
✓ Launch Engineering Client	04-03-2025 10:19:34	00:01:05
✓ Logout - click under dropdown in EC	04-03-2025 10:20:40	00:00:05
✓ Click - Login - from dropdown and enter valid username, password and click login	04-03-2025 10:20:45	00:00:26
✓ Creation of System - By Right clicking on System Explorer	04-03-2025 10:21:12	00:00:13
✓ Rename of System as Control_Test	04-03-2025 10:21:25	00:00:05
✓ Creation of System - By Right clicking on System Explorer	04-03-2025 10:21:30	00:00:14
✓ Navigation To Application Explorer(AE)	04-03-2025 10:21:45	00:00:14
✓ Expand all Hirarchy of AE Panes	04-03-2025 10:21:59	00:05:02
✓ Search Template in Template browser and Drag and drop from template browser to application browser	04-03-2025 10:27:01	00:02:57
✓ Expand Asset workspace	04-03-2025 10:29:59	00:01:07
✓ Replace template - Select template to replace template from Application browser	04-03-2025 10:31:06	00:00:22
✓ Replace template - Click on OK in replace template window	04-03-2025 10:31:28	00:00:13
✓ Import - RightClick on the system_1 and select Import from the context menu	04-03-2025 10:31:42	00:00:10
✓ import - Enter the filelocation and select the one instances file to be imported click on OK in popup	04-03-2025 10:31:59	00:00:33

✓ import - Click on Update all button from resolve conflict pop up and ok	04-03-2025 10:32:38	00:00:16
✓ import - Verify Template added in AE	04-03-2025 10:32:54	00:00:03
✓ Rename instance - invalid 1234	04-03-2025 10:32:57	00:00:14
✓ Delete instance - from context Menu	04-03-2025 10:33:17	00:00:21
✓ Search MotorGP in Template browser and Drag and drop from template browser to application browser	04-03-2025 10:33:44	00:00:40
✓ Modify the properties of Instances - Edit Instance	04-03-2025 10:34:24	00:00:31
✓ Modify the properties of Instances - Save	04-03-2025 10:34:56	00:00:01
✓ Modify the properties of Instances - Close Instance	04-03-2025 10:34:57	00:00:31
✓ Inspect instance - Rclick on the inspect instance and verify the tab and click on ok	04-03-2025 10:35:29	00:00:11
✓ Create 2 folders in Application browser	04-03-2025 10:35:41	00:00:10
✓ Create 2 folders in Application browser	04-03-2025 10:35:51	00:00:11
✓ Create 2 folders in Application browser	04-03-2025 10:36:03	00:00:11
✓ Copy and paste the Instance - copy instance from context menu and paste	04-03-2025 10:36:14	00:00:24
✓ Export instance - Rclick root node (folder), Export as XML, verify XML file	04-03-2025 10:36:39	00:00:24
✓ Navigation To Topology Explorer(TE) - Main Tool Bar	04-03-2025 10:37:04	00:00:18
✓ Navigate to Project Explorer by shortcut (ALT+P)	04-03-2025 10:37:22	00:00:06
✓ Navigate to System Explorer (SE)	04-03-2025 10:37:28	00:00:09
✓ Navigation to Global template Explorer(GTE)	04-03-2025 10:37:37	00:00:12
✓ Navigation to Content Repository(CR)	04-03-2025 10:37:50	00:00:10
✓ Navigate to System Explorer (SE)	04-03-2025 10:38:00	00:00:09
✓ Navigation To Project Explorer(PE) - Main Tool Bar by selecting system	04-03-2025 10:38:09	00:00:09
✓ Create a Control Project for M580_Standalone	04-03-2025 10:38:23	00:00:11
✓ Double Click on Container sections	04-03-2025 10:38:44	00:00:04
✓ Assign Instance from system to different Containers in PE	04-03-2025 10:38:58	00:00:25
✓ Trying to Unassign Before Generating	04-03-2025 10:39:27	00:00:23
✓ Creating Multiple section and verify	04-03-2025 10:39:51	00:01:57
✓ Delete two Sections	04-03-2025 10:41:48	00:00:50
✓ Moving the Instance from One section to Another without Generating	04-03-2025 10:42:39	00:00:42
✓ Generate from Containers pane	04-03-2025 10:43:21	00:00:30



Business Impact **Delivered**



Accelerated Time-to-Market

- Automation of critical smoke tests enabled daily confidence builds.
- Reduced regression test cycle from **3 weeks to 4.5 hours** with automation and distributed execution.



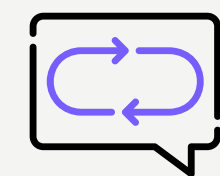
Improved Quality & Coverage

- End-to-end mapping of test cases to scripts ensured better coverage.
- Achieved **40% code coverage** in Phase 1.
- Early defect detection through multi-system execution.



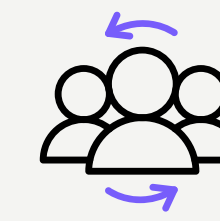
Reduced Testing Costs

- Python-based automation and low-code tools cut scripting and maintenance effort by **50%**.



Actionable Feedback Loop

- Integrated dashboards and detailed reports led to faster issue resolution.
- Enabled data-driven decision-making for QA and release teams.



Decreased People Dependency

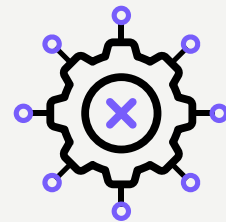
- Simplified scripting and reusable components democratized automation.



Why This Matters to You



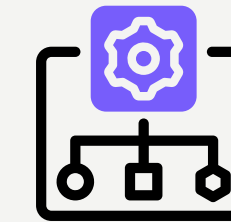
We have **domain experience** and understand the specific needs of Industrial Automation and DCS systems, such as real-time responsiveness, hardware-software interactions, and safety validation.



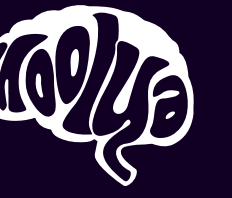
Our solution is not just technical—it's **tailored to your business context**.



By replicating our proven strategy, we can help you achieve **faster automation ROI**, better release predictability, and **higher system reliability**.



Our modular, maintainable framework can be adapted to your current tools and teams with minimal disruption.



Thank You!